Scalability & Future Plan

Date	30 June 2024
Team ID	SWTID-2026-4836
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine.
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only a limited agricultural dataset.	May reduce prediction accuracy for other regions and crops.	High
2	Application is currently designed for local deployment only.	Cannot support a large number of simultaneous users.	Medium
3	No user authentication and cloud database integration.	Limited security and scalability.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of users on a local server.	Deploy on cloud platforms such as Render, AWS, or Azure with load balancing.
2	Data Storage	Uses local CSV dataset and model files.	Integrate MySQL or MongoDB for large-scale agricultural data storage.
3	Performance	Predictions are generated using a locally stored model.	Optimize model performance and use caching techniques for faster predictions.
4	Security	Basic input validation is implemented.	Add user authentication, HTTPS, and secure API integration.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment and online hosting	2 Months	Improve accessibility and support more users.
Phase 2	Real-time weather API integration	3 Months	Provide more accurate crop recommendations.
Phase 3	Mobile application development	4 Months	Enable farmers to access the system through smartphones.
Phase 4	Advanced AI models and multilingual support	6 Months	Increase prediction accuracy and usability for farmers in different regions.